Data Mining

k-Nearest Neighbor Classifiers

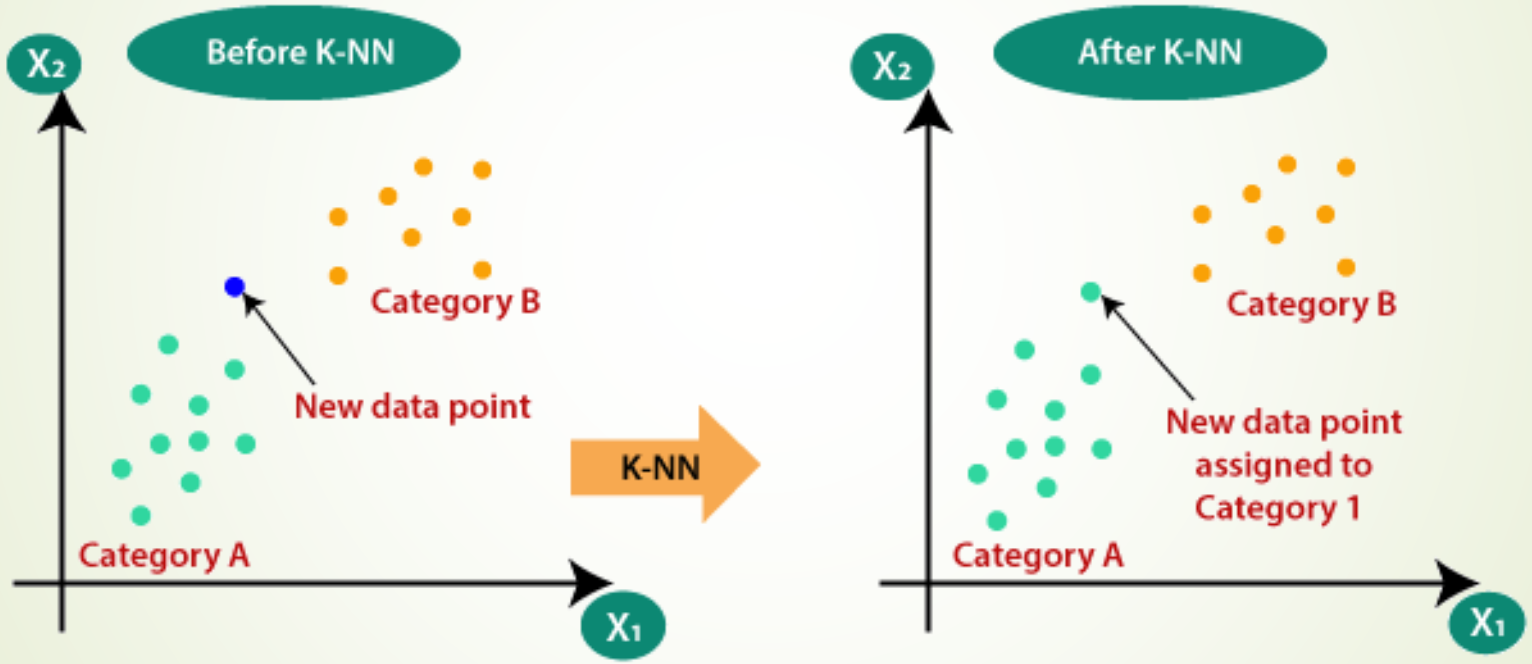
Introduction

- Instance-based (lazy) learning algorithm
- No explicit training model → classification done at query time
- Uses distance metrics to compare similarity
- Popular for simple yet effective classification tasks

Working Principle

- Store all training instances
- Given a new sample, compute distances to all training points
- Select the k nearest neighbors
- Predict class by majority vote among neighbors

Working Principle



Choice of k

- Small k → sensitive to noise, may overfit
- Large k → smoother decision boundary, may underfit
- Rule of thumb: odd k values for binary classification
- Common practice: tune k using cross-validation

Distance Metrics

- Euclidean Distance (most common)
- Manhattan Distance
- Minkowski Distance
- Cosine Similarity (for text/vector data)

Advantages

- Simple, intuitive, easy to implement
- No training phase → flexible
- Naturally handles multi-class problems
- Performs well with enough representative data

Disadvantages

- Computationally expensive at prediction time
- Requires storing the entire dataset
- Sensitive to irrelevant features & scaling
- Poor performance on high-dimensional data (“curse of dimensionality”)